RedSOC

Adversarial Evaluation Framework for LLM-Integrated Security Operations Centers

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github.com/krishnakaanthreddyy1510-cell/RedSOC | doi.org/10.6084/m9.figshare.32016498

① ADVERSARIAL ATTACK SURFACE — THREE ATTACK CLASSES

INDIRECT PROMPT INJECTION

Adversarial instructions embedded in threat intel documents retrieved by RAG pipeline.
No SOC infrastructure access required.
Exploits document ingestion trust boundary.

Lowest attacker capability

100% ASR | SD = 0% | 10 runs

CORPUS POISONING

Malicious docs injected into FAISS vector knowledge base via trusted ingestion channels.
Satisfies retrieval + generation conditions.
Persistent — affects all future queries.

Persistent threat

80% ASR | SD = 0% | 10 runs

DIRECT PROMPT INJECTION

Override phrases appended to analyst queries via workstation access or social engineering.
Session-level misdirection of AI guidance.
Partial resistance from model safety training.

Highest attacker capability

60% ASR | SD = 0% | 10 runs

Attack vectors enter pipeline

② SOC-RAG PIPELINE

ANALYST QUERY INPUT

Security analyst submits natural language query to AI SOC assistant.
Direct injection attacks target this input stage directly.

SEMANTIC RETRIEVAL — FAISS VECTOR STORE

Query is embedded via sentence-transformers (all-MiniLM-L6-v2) and compared against knowledge base using cosine similarity. Top-k documents retrieved.
Corpus poisoning + indirect injection exploit this stage — malicious docs satisfy the retrieval condition. This is the primary attack surface.

LLM RESPONSE GENERATION

Retrieved docs + query passed to language model. Response grounded in retrieved context — including any adversarial content.

Query + retrieved docs + response to detection

③ THREE-MECHANISM DETECTION LAYER — NO MODEL INTERNALS REQUIRED

MECHANISM 1 — SEMANTIC ANOMALY DETECTION

Cosine similarity between query embedding and each retrieved document embedding.
Adversarially crafted docs exhibit lower semantic alignment with legitimate queries.

```
SAD = 1[ avg cos(e_q, e_di) < t=0.7 ]
```

MECHANISM 2 — PROVENANCE TRACKING

Maintains whitelist of trusted source IDs.
Any retrieved doc from unwhitelisted source flagged immediately — regardless of content.
Deterministic. Not threshold-based.

```
PT = 1[ exists i : source_i not in S ]
```

MECHANISM 3 — RESPONSE CONSISTENCY CHECKING

Semantic alignment between generated response and retrieved document context.
Responses steered by injected instructions diverge semantically from legit context.

```
RCC = 1[ cos(e_response, avg e_di) < t ]
```

Three signals to unified verdict engine

④ UNIFIED VERDICT ENGINE + BENCHMARK RESULTS — 15 SCENARIOS · 10 INDEPENDENT RUNS

UNIFIED THREAT VERDICT — AGGREGATION OF THREE MECHANISMS

Threshold: >= 2 mechanisms triggered = HIGH -> BLOCK + ALERT ANALYST
1 mechanism triggered = MEDIUM -> FLAG FOR ANALYST REVIEW
0 mechanisms triggered = LOW -> PASS
No model internals required. Works with GPT-4o, Claude, Gemini, any hosted API.

100%	**80%**	**60%**	**100%**
Indirect Injection ASR	Corpus Poisoning ASR	Direct Injection ASR	Detection Rate DR